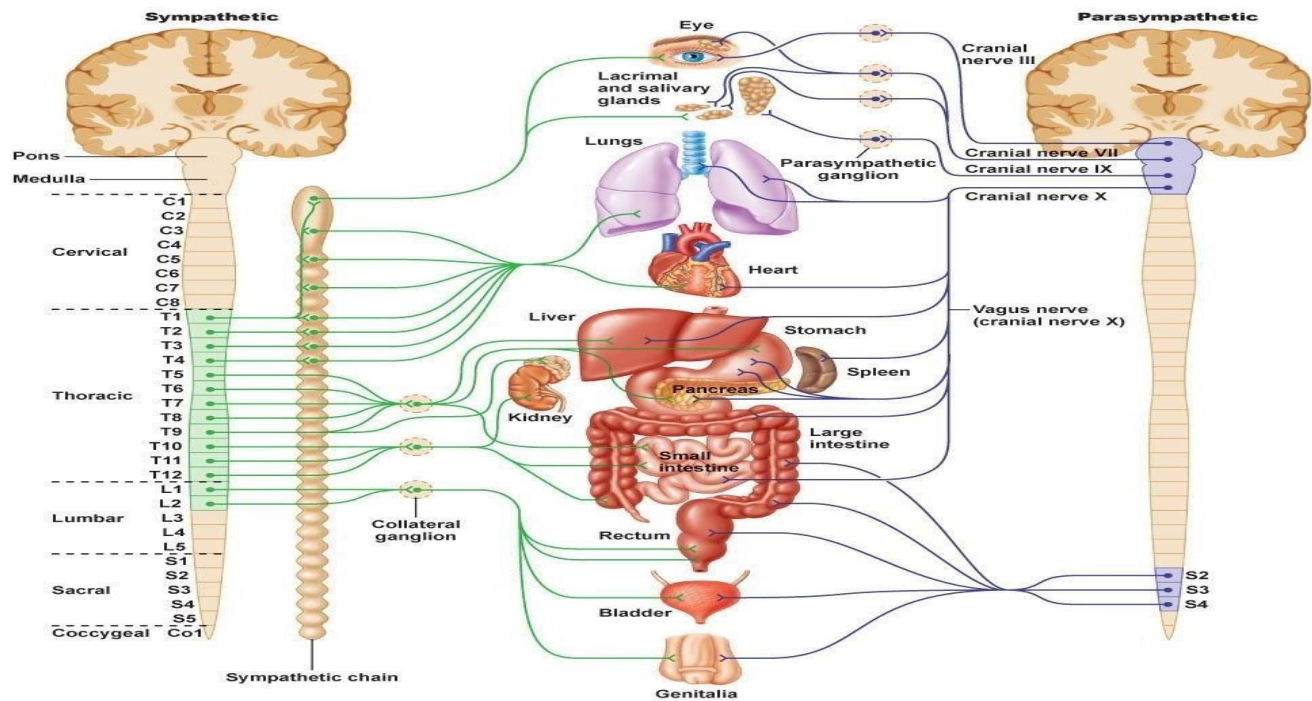
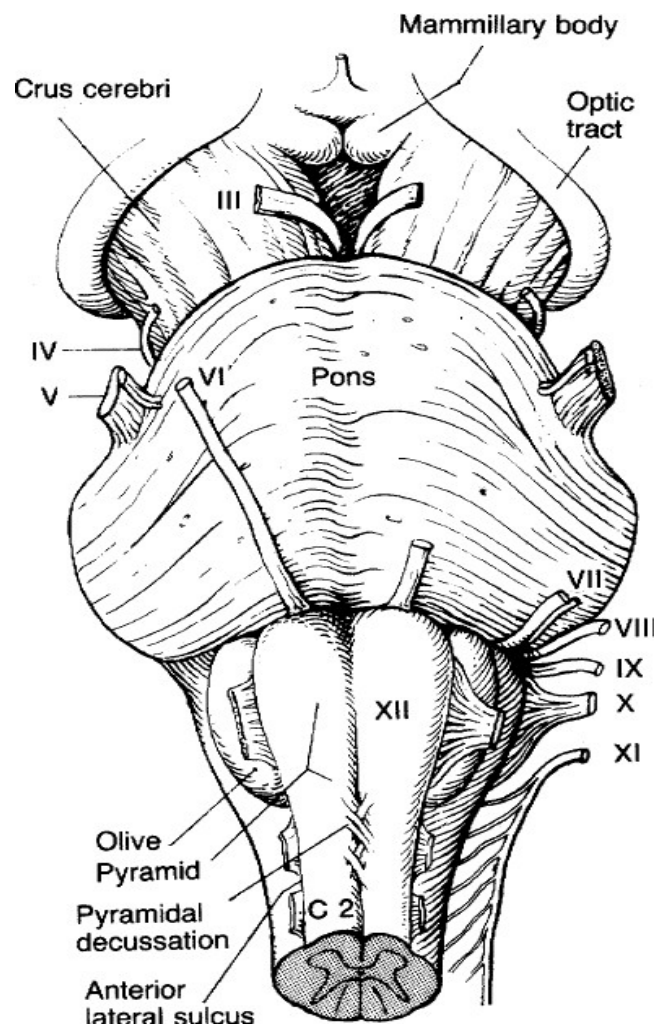
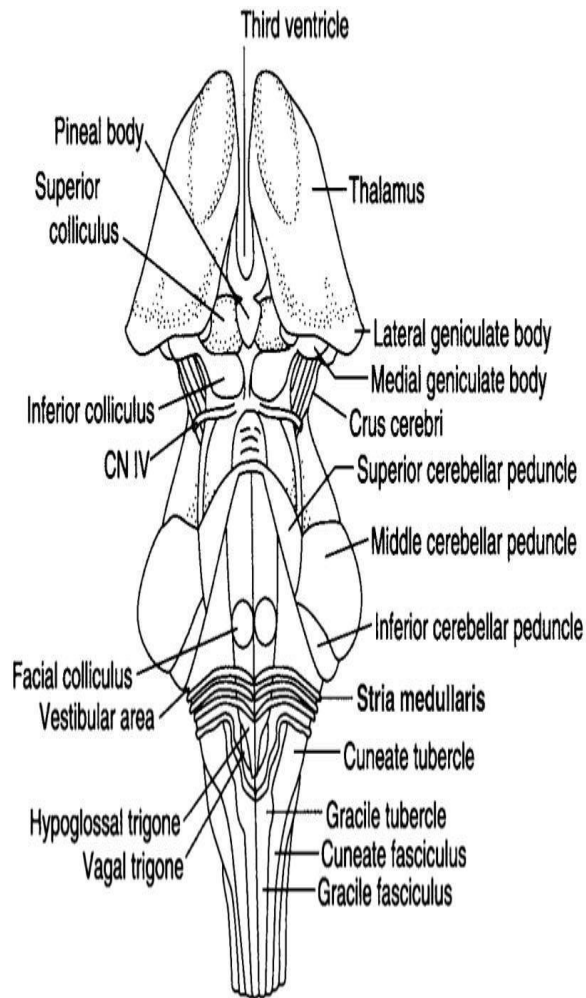




Lecture (13): Brainstem

- The brainstem is the distal part of the brain that connects the cerebrum, diencephalon and cerebellum with the spinal cord.
- It consists of:
 - 1- The **midbrain** is the widest and most superior segment.
 - 2- The **pons** is the middle segment of the brainstem.
 - 3- The **medulla oblongata** is the narrowest and distal part.
- It extends from just above the aperture in the tentorium cerebelli to just below the foramen magnum.
- The cerebellum projects from its dorsal surface.
- The brainstem is widest at its proximal end and becomes narrower toward the distal end.





Medulla oblongata:

- The medulla oblongata (**MO**) is the narrowest and most caudal part of the brainstem. It is a funnel-like structure.
- The **MO** extends from the decussation of the great pyramids at foramen magnum inferiorly to the inferior pontine sulcus superiorly.
 - The **medullary striae** of the fourth ventricle posteriorly separate back of **MO** from back of **pons**.
- **Parts of MO:**
 - **Closed medulla** (lower half): it contains a central canal.
 - **Open medulla** (upper half): related posteriorly to 4th ventricle.

External features of medulla

Ventral aspect:

- ▣ **Anterior median fissure** divides the medulla longitudinally into Rt & Lt halves.

▣ **Three elevations** on each side of the anterior median fissure:

- **Pyramid:** close to midline. It is formed of corticospinal (pyramidal) tract.
- **Olive:** lateral to the pyramid. It is formed by inferior olivary nucleus.
- **Inferior cerebellar peduncle:** lateral to olive, connecting medulla to cerebellum.

▣ **Exit of lower 4 cranial nerves:**

- **Hypoglossal (CNXII)** nerve appears in antero-lateral sulcus between pyramid and olive.
- **Glossopharyngeal (CNIX), Vagus (CNX) and Cranial root of accessory (CNXI)** nerves from **cranial to caudal** appear in postero-lateral sulcus between olive and inferior cerebellar peduncle.

Dorsal aspect:

▣ **Closed medulla (lower part):**

It carries **3 elevations** on each side of the posterior median sulcus: from medial to lateral:

- **Gracile tubercle** (formed by gracile nucleus) and **gracile tract**.
- **Cuneate tubercle** (formed by cuneate nucleus) and **cuneate tract**.
- **Tuberculum cinereum:** formed by spinal nucleus of trigeminal.

▣ **Open medulla (upper part):**

- It forms the lower part of **floor of 4th ventricle**.
- It shows **3 trigones** separated by inferior fovea:
- **Hypoglossal trigone:** medial, formed by hypoglossal nucleus.
- **Vagal trigone:** middle, formed by dorsal motor vagal nucleus.
- **Vestibular trigone:** lateral, formed by vestibular nuclei.

Internal Structure of medulla

- The **corticospinal tract** occupies and forms the **pyramid**. Inferiorly, about **85%** of corticospinal tract cross to opposite side forming the **pyramidal (motor) decussation** at lower part of closed medulla.
- **Sensory decussation** is formed by the crossing **internal arcuate fibers**, which are the axons of gracile and cuneate nuclei. It is located **above** pyramidal decussation. The decussated fibers ascend as the medial lemniscus.
- **Gracile nucleus:**
 - It is located on the **back of closed medulla**, close to the middle line.
 - It receives **axons of gracile tract**.
 - Its axons form **Internal arcuate fibers** which cross to opposite side to form

medial lemniscus.

- It carries **proprioception and fine touch** from the lower 1/2 of the body.

o Cuneate nucleus:

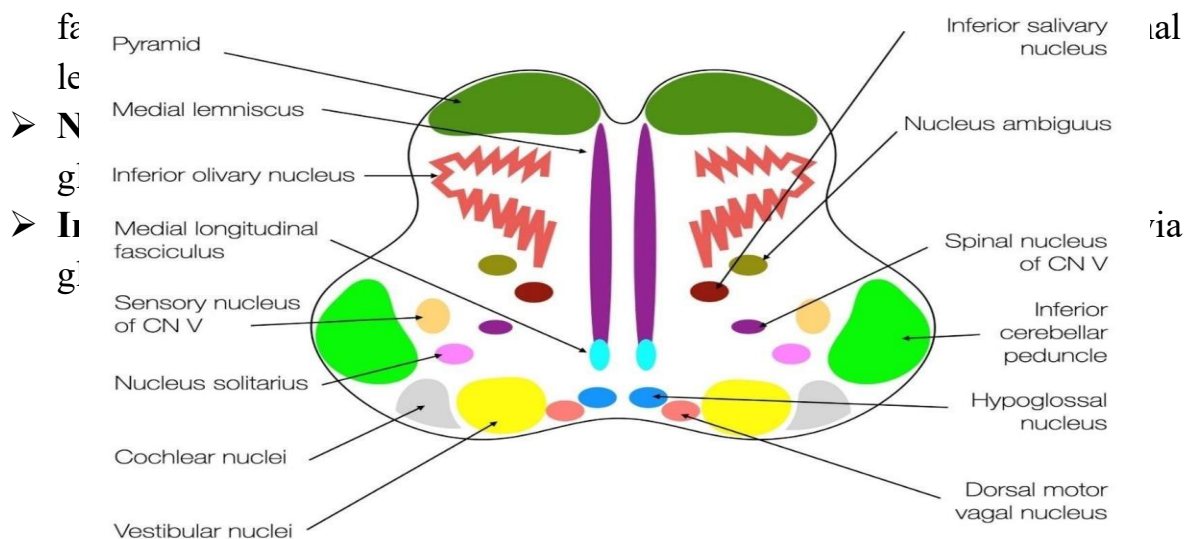
- It is located on the **back of closed medulla**, lateral to gracile nucleus.
- It receives **axons of cuneate tract**.
- Its axons form **Internal arcuate fibers** which cross to opposite side to form **medial lemniscus**.
- It carries **proprioception and fine touch** from the upper 1/2 of the body.

o Inferior olivary nucleus:

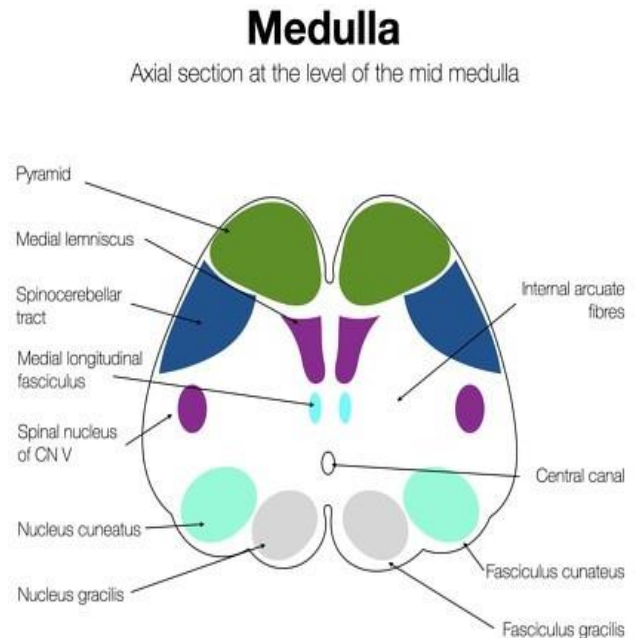
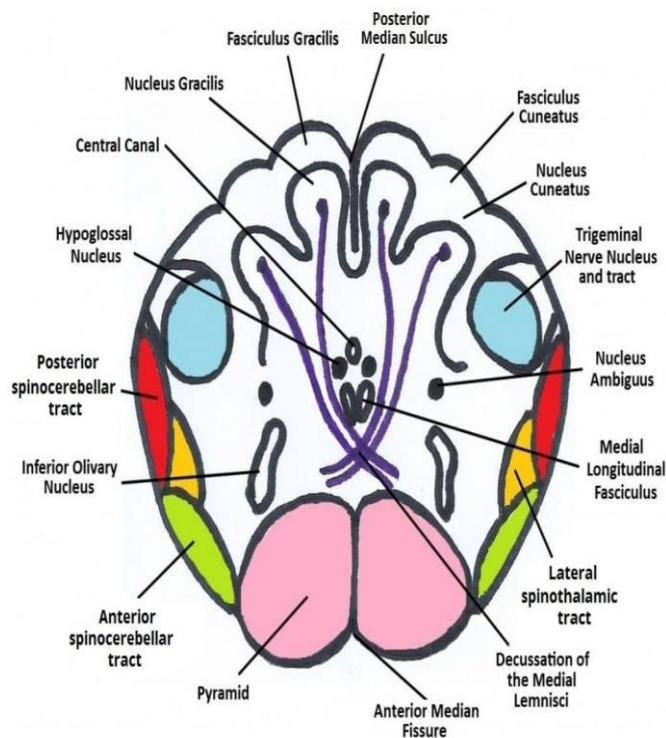
- It is located **lateral to pyramid in open medulla**.
- It is a **crenated C-shaped** with an open end facing medially.
- Receives information via spinal cord & cerebral cortex.
- Its axons cross the midline to form **olivo-cerebellar tract** that enters cerebellum via inferior cerebellar peduncle.
- Its function is **motor coordination and learning** through its connection with cerebellum.

o Cranial nerve nuclei in medulla:

- **Hypoglossal nucleus:** In the floor of 4th ventricle close to the midline, it gives rise to hypoglossal nerve which supplies all muscles of tongue except palatoglossus.
- **Nucleus ambiguus:** In the medulla. It gives **motor fibers to 9th, 10th, 11th cranial nerves** to supply all muscles of pharynx, larynx & soft palate except tensor palati.
- **Dorsal motor nucleus of vagus:** (parasympathetic nucleus). Lies lateral to hypoglossal nucleus.
- **Spinal nucleus of trigeminal:** Carries pain and temperature sensations from the



Cross section in open medulla



PONS

- The pons is the middle segment of brainstem and is located between midbrain superiorly and medulla inferiorly.
- The pons is anterior to the cerebellum like a bridge connecting two cerebellar hemispheres.

External features of pons

Ventral aspect:

- The anterior surface is convex from side to side.
- It shows transverse pontine fibers that converge laterally to form the middle cerebellar peduncle.
- There is a midline **basilar groove**, which lodges the **basilar artery**.
- **Trigeminal nerve** emerges on the anterolateral surface of the pons. Each nerve consists of a **smaller motor root**, and a **larger sensory root**.
- **Abducent nerve** appears in the groove between the pons and the pyramid of medulla.
- **Facial & Vestibulocochlear nerves** exit in cerebellopontine angle.

- **Cerebellopontine angle** is the space at the junction between pons, medulla and cerebellum. It contains: **Facial and vestibulocochlear nerves, Anterior inferior cerebellar artery, Choroid plexus & foramen of Luschka of the 4th ventricle.**

Dorsal aspect:

- It forms the **upper half of the floor of the 4th ventricle.**
- It is separated from back of **open medulla** by **stria medullaris** (transverse fibers from **arcuate nuclei**).
- Pons is bounded laterally by the superior cerebellar peduncles.
- **Medial eminence** is an elongated elevation, produced by **abducent nucleus.**
- **Facial colliculus** is an expanded inferior end of the medial eminence. which is produced by the root of the facial nerve **winding (encircling)** around the abducent nucleus.
- Lateral to the sulcus limitans is the **area vestibuli.**

Internal Structure of pons

- ▣ It is divided into two parts:
 - Anterior part is called the basis pontis.
 - Posterior part is called the tegmentum.
- ▣ **Basis pontis** contains:
 - Pontine nuclei.
 - Transverse pontocerebellar fibres cross to opposite side to the middle cerebellar peduncle.
 - Corticopontine fibres descend from cerebral cortex to pontine nuclei.
 - Bundles of pyramidal tract.
- ▣ **Tegmentum** contains:
 - Nuclei of the middle four cranial nerves.
 - Ascending sensory tracts (**Lemnisci**).
 - Pontine reticular formation.

Nuclei of middle four cranial nerves:

- ▣ **Nuclei of trigeminal nerve:**
 - **Motor nucleus of trigeminal:** lies at mid-level of pons, it supplies 4 muscles of mastication, mylohyoid, anterior belly of digastric, tensor palati and tensor tympani.
 - **Main sensory nucleus:** Lies lateral to motor nucleus, It receives touch sensation from the same side of face through central processes of **trigeminal ganglion.**
 - **Spinal nucleus of trigeminal:** It extends downward in **medulla**, it receives pain and temperature from the same side of the face.

- **Mesencephalic nucleus:** Lies in the **midbrain** and extends downward into pons. It is a **1st order neuron** for **proprioception** from muscles of mastication & TMJ.

■ Motor nucleus of abducent nerve.

Lies in the floor of 4th ventricle close to the midline. It innervates **lateral rectus muscle**.

■ Nuclei of facial nerve.

- **Motor nucleus:** It is located in the lower part of pons and the facial nerve fibers make a loop around abducent nucleus to form the **facial colliculus**.
- **Upper half of facial nucleus** receives **bilateral corticobulbar fibers** and supply upper facial muscles.
- **Lower half of facial nucleus** receives **contralateral corticobulbar fibers** and supply lower facial muscles.
- **Nucleus solitarius:** Lies in medulla, it receives taste sensation from the anterior **2/3** of the tongue through chorda tympani.
- **Superior salivatory nucleus:** Parasympathetic nucleus located in lower pons. It supplies submandibular and sublingual salivary glands.

■ Nuclei of vestibulocochlear nerve.

- **Four vestibular nuclei:**
 - The **superior & Lateral** nuclei (located in lower pons).
 - **Inferior & Medial** nuclei (located in medulla) lie in the vestibular area beneath the floor of the 4th Ventricle.
 - They receive from **Membranous labyrinth** (maculae utricle & saccule in vestibule and 3 cristae ampullares in 3 semicircular canals via vestibular nerve).
- **Superior olivary nucleus:**
 - It lies in lower pons.
 - It is 1st neuron in central auditory system for **binaural hearing**.
 - It is important for **localization & direction of sounds**.
- **Ventral & dorsal cochlear nuclei:**
 - They are in **lower pons**.
 - They **receive fibers of cochlear nerve** transmit hearing sensation from organ of Corti.
 - Axons of ventral cochlear nucleus cross midline forming **trapezoid body** that relay in **superior olivary nucleus and lateral lemniscus**.

Four sensory lemnisci:

1- Lateral lemniscus:

- The most lateral lemniscus. It is an ascending brainstem auditory pathway.
- It carries hearing sensation from both ears.
- It originates from axons of contralateral cochlear nuclei and from superior olivary nucleus. It ends in inferior colliculus of midbrain and medial geniculate body of thalamus.

2- Spinal lemniscus:

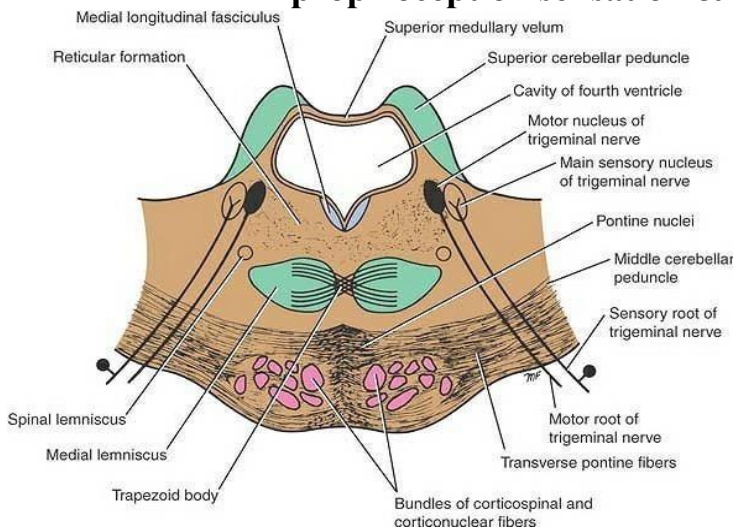
- Just medial to lateral lemniscus. It originates from axons of substantia gelatinosa of Rolandi & nucleus proprius of spinal cord that ascend as ventral & lateral spino-thalamic tracts. It ends in PLVNT of thalamus.
- It carries pain, temperature & crude touch from opposite side of the body.

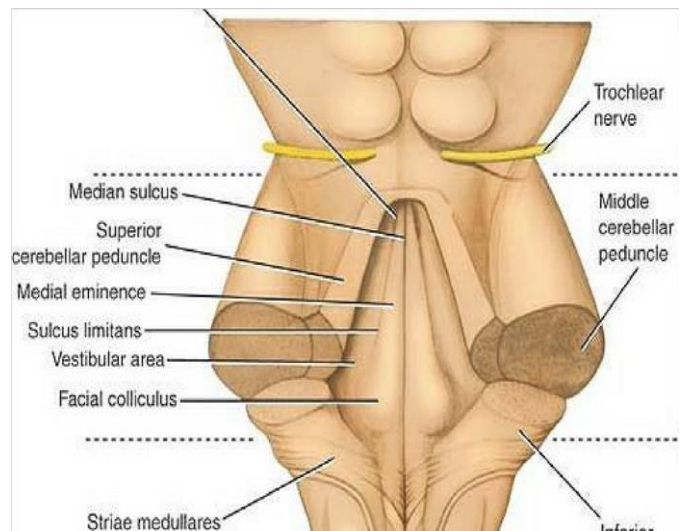
3- Trigeminal lemniscus:

- just medial to spinal lemniscus.
- It **originates from axons of Spinal nucleus & main sensory nucleus of trigeminal**. It ends in **PMVNT** of thalamus.
- It carries general somatic sensation from the opposite side of the face & head.

4- Medial lemniscus:

- **The most medial lemniscus.**
- It **originates** from axons of gracile and cuneate nuclei. It ends in **PLVNT** of thalamus. It carries **proprioception sensation & fine touch** from opposite side of body.

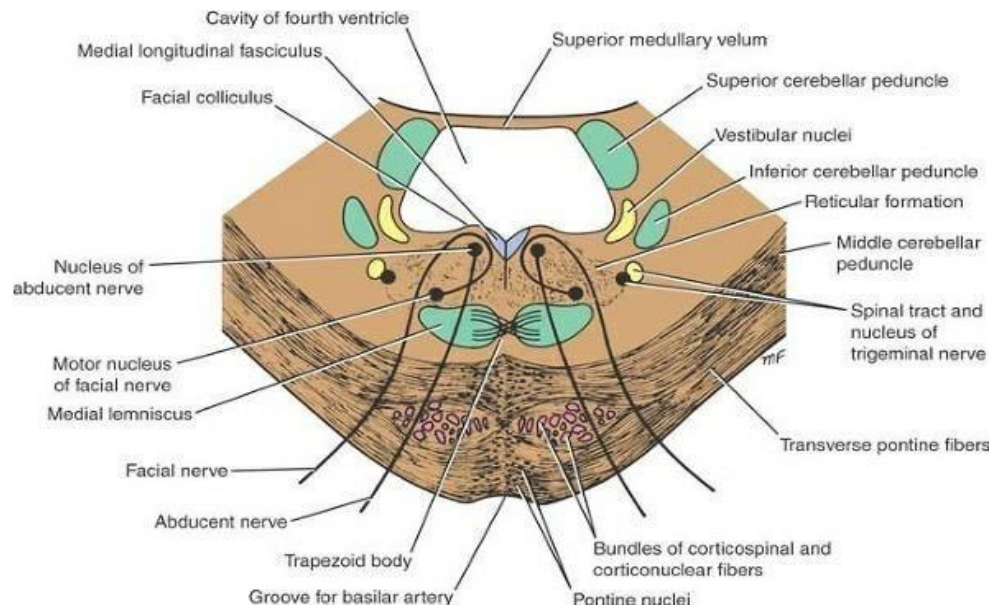




Cross section mid-pons

back of pons

Cross section at lower part of pons



Midbrain

- ❑ The midbrain is about (2 cm) in length and connects the pons and cerebellum with the forebrain.
- ❑ It ascends through the opening in the tentorium cerebelli.
- ❑ It is traversed by a narrow channel, the cerebral aqueduct of sylvius, which connects the 3rd & 4th ventricles.
- ❑ It is connected to cerebellum by two superior cerebellar peduncles.

External features of midbrain

Ventral aspect:

- The midbrain consists of right and left halves, each half forming a **cerebral peduncle**.
- There is a deep depression in between two cerebral peduncles called the **interpeduncular fossa**.
- The **oculomotor nerve** emerges from interpeduncular fossa and medial to the crus cerebri.

Dorsal aspect:

- There are four colliculi (corpora quadrigemina), which are divided into superior and inferior pairs by a vertical and a transverse groove.
- The **superior colliculi** are centers for visual reflexes.
- The **inferior colliculi** are lower auditory centers.
- The **trochlear nerves** emerge in the midline below the inferior colliculi, and winds around lateral aspect of midbrain.

Lateral aspect:

- The **superior brachium** passes from the superior colliculus to the lateral geniculate body.
- The **inferior brachium** connects the inferior colliculus to the medial geniculate body.
- Structures **wind around lateral side of midbrain**: Posterior cerebral arteries, Superior cerebellar arteries, Trochlear nerve and Optic tract.

Internal Structure of Midbrain:

- o The midbrain is divided by the cerebral aqueduct of Sylvius into two portions:
 - **Ventral portion (Cerebral peduncle)**: which is divided into **3** parts:
 - **Crus cerebri** (basis pedunculi): anterior part.
 - **Substantia nigra**: middle part.

- **Tegmentum:** posterior part.
- **Dorsal portion (tectum).**

Cerebral Peduncle:

- **Crus cerebri:** It consists of the following fibers:
 - **Medial 1/5:** frontopontine fibers & medial corticobulbar tract.
 - **Middle 3/5:** corticospinal tract.
 - **Lateral 1/5:** Temporo-pontine fibers & lateral corticobulbar tract

Substantia nigra

- It is a large, pigmented motor nucleus situated between the tegmentum and the crus cerebri and is **found throughout the midbrain.**
- The nucleus is composed of multipolar neurons that possess granules of melanin pigment.
- It is one of **extrapyramidal centers.**
- It is formed of:
 - **Pars reticulata** (anterior) secretes GABA.
 - **Pars compacta** (posterior) secrete dopamine.

Functions:

- The substantia nigra is concerned with muscle tone and is connected to the cerebral cortex, spinal cord, and basal nuclei.
- Regulates function of basal ganglia.
- Secretes dopamine which is a pleasure hormone.

Nuclei of oculomotor nerve:

1- Motor nucleus:

- It is situated at **level of superior colliculus.**
- Its fibers traverse anteriorly through the **red nucleus** to emerge on the **medial side of crus cerebri**, in the **interpeduncular fossa.**
- It supplies **all extra-ocular muscles**, **except lateral rectus & superior oblique muscles.**

2- Edinger-Westphal nucleus:

- It is parasympathetic nucleus that lies close to oculomotor nucleus.
- Its axons run out with the third nerve and relay in ciliary ganglion, then supply **sphincter pupillae & ciliary muscles.**

Trochlear nucleus:

- It is situated **at level of inferior colliculus.**

- Its axons cross to the opposite side and leave the midbrain just below inferior colliculi.
- It supplies **superior oblique muscles** of the eye.

Red nucleus

- The red nucleus is a **rounded mass of gray matter** situated in **tegmentum** of midbrain between the cerebral aqueduct and the substantia nigra at level of superior colliculus.
- Its reddish hue is due to its vascularity and the presence of iron-containing pigment in their cytoplasm.
- It is an **extra pyramidal center**.
- **Input:** Cerebral cortex, Cerebellum and Basal ganglia.
- **Output:** Spinal cord through rubrospinal tract, Motor nuclei of cranial nerves and Thalamus.
- **Functions:**
 - It controls gross movements and facilitates tone of flexors of upper limbs.
 - It mediates motor impulses from cerebral cortex and cerebellum to thalamus.
 - Controls crawling in babies.

Tectum of midbrain:

- The tectum is the posterior part of midbrain behind the cerebral aqueduct; it has four small surface swellings; these are two superior and two inferior colliculi.

Superior colliculus

- Superior colliculus is a **large, laminated nucleus**.
- It is connected to the **lateral geniculate body** by the **superior brachium**.
- It relates to the opposite one by the **posterior commissure**.
- It receives **afferent** fibers from the optic nerve, the visual cortex, the spinal cord (spinotectal tract) and Inferior colliculus of the same side.
- The **efferent fibers** form the tectospinal & tecto-bulbar tracts.
- It is responsible for the **reflex movements of the eyes, head, and neck in response to visual stimuli**.
 - **Control conjugate eye movement & fix object on fovea (sharpness of vision).**
- The pupillary light reflexes involve the **pretectal nuclei**, which lies just cranial to the superior colliculi.

Inferior colliculus

- The inferior colliculus consists of a large nucleus of gray matter.
- It forms a part of the **central auditory pathway**.

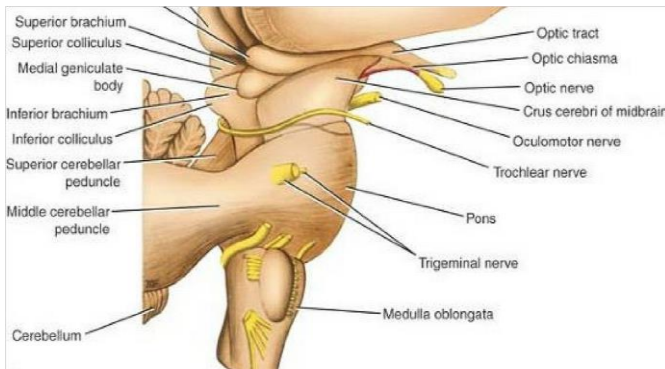
- It receives many of the terminal fibers of the lateral lemniscus & inferior colliculus of the opposite one. It sends to the **medial geniculate body** through the **inferior brachium** and to **superior colliculus** of the same side.
- It is **auditory reflex center**.

Interpeduncular fossa

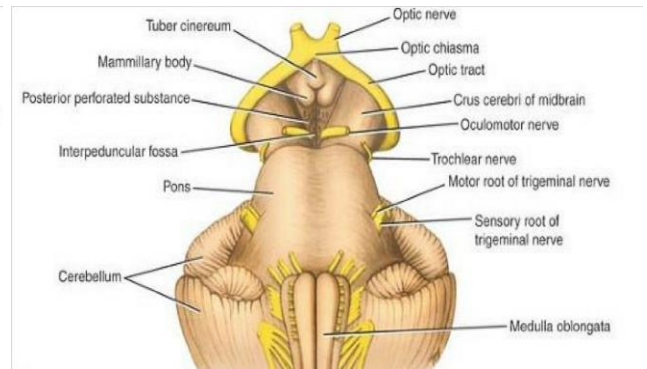
- Shape: Diamond shaped depression at the base of brain in front and between the 2 cerebral peduncles.
- Boundaries:
 - Anteriorly: optic chiasma.
 - Posteriorly: upper border of the pons.
 - Anterolaterally: optic tracts.
 - Posterolaterally: cerebral peduncles.
 - Floor: - Posterior perforated substance: gray brain matter has many small holes that allow blood vessels to pass through to the brain, 2 mammillary bodies: part of the limbic system, The tuber cinereum & Pituitary stalk (infundibulum): connects the gland to hypothalamus.

■ Contents:

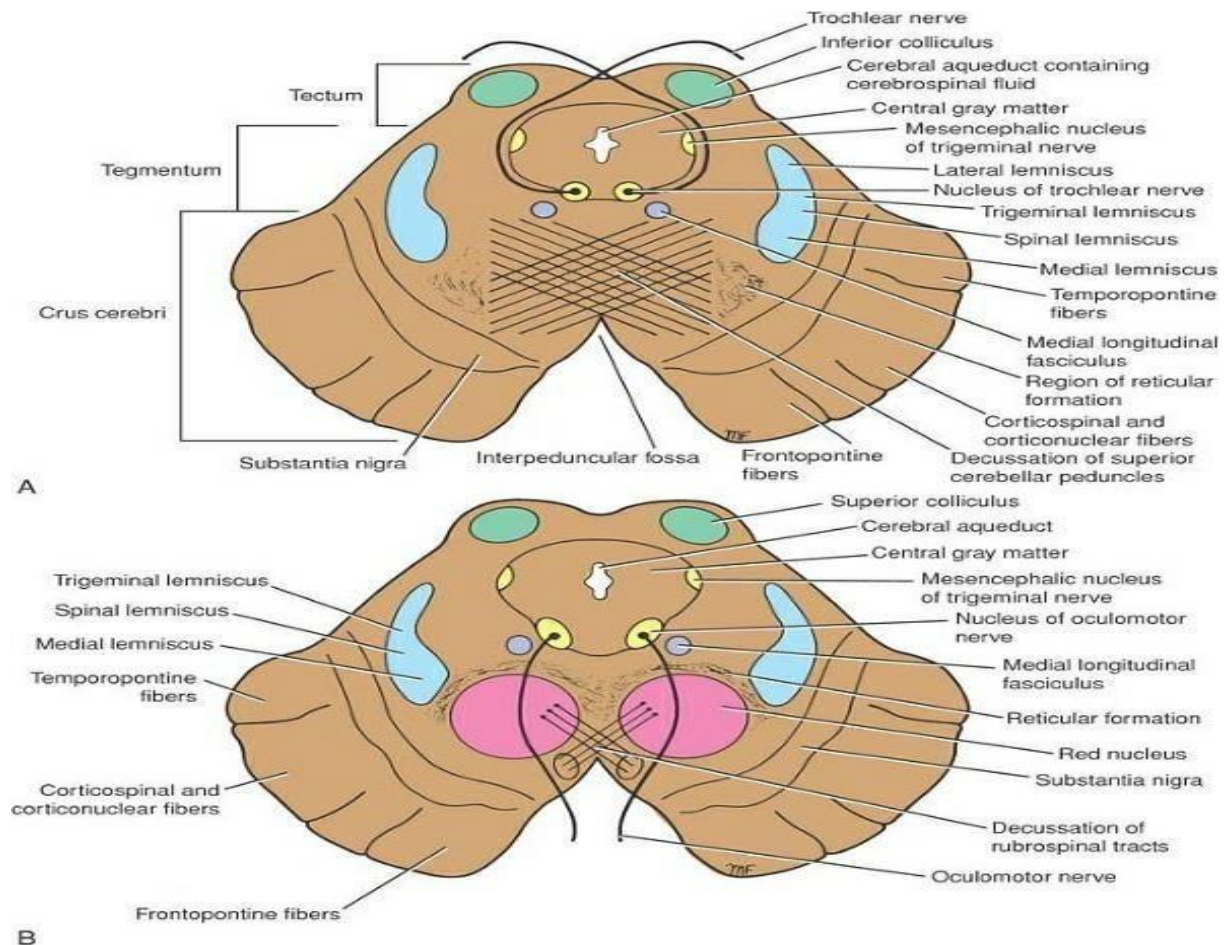
- 1- Interpeduncular cistern: dilatation in subarachnoid space.
- 2- Circulus of Willis.
- 3- Oculomotor nerve.



Lateral aspect of midbrain



Ventral aspect of midbrain



Transverse sections of midbrain at level of inferior colliculus (A) & superior colliculus (B)

Lecture (14): cerebrum I

- The cerebrum is the largest part of the brain, situated in the cranial cavity.
- It is divided into two parts: **diencephalon** and **cerebral hemispheres**.
- The **two cerebral hemispheres** are separated by a deep midline **longitudinal fissure** which is occupied by **falx cerebri** and **two anterior cerebral arteries**.
- The two cerebral hemispheres are connected by the **corpus callosum** across the midline.
- The cerebral hemispheres are separated horizontally from the cerebellum by the **tentorium cerebelli**.
- The surface of each cerebral hemisphere is thrown into a complicated series of tortuous folds called **gyri**, which are separated from each other by **sulci** or **fissures**.
- **Lobes of cerebrum**: the cerebral hemisphere is divided into **4 lobes**:
 - **Frontal lobe** lies in front of the central sulcus and above the lateral sulcus.